

DFx - CFU Playground

Pynq flow

Top Bitstream Initialisation

```
In [62]: from pynq import Overlay , Bitstream
import time
overlay = Overlay("Top.xsa")

config_to = overlay.axi_gpio_config_to
decouple = overlay.axi_gpio_decouple
decouple_status = overlay.axi_gpio_decouple_status
to_reconfig = overlay.axi_gpio_to_reconfig
perf_counter = overlay.axi_gpio_perf_counter
perf_rst = overlay.axi_gpio_perf_rst
```

Functions Used

```
In [63]: pr_files = {
    0: {
        "mode" : "Donut",
        "file" : "cfu_system_Cfu_cfu_inst_Donut_partial.bit"
    },
    1: {
        "mode" : "Shift_Mul",
        "file" : "cfu_system_Cfu_cfu_inst_Shift_Mul_partial.bit"
    }
}

def reconfigure(ID):
    if(ID in pr_files.keys()):
        add_pb = Bitstream(pr_files[ID]["file"], None, True)
        add_pb.download()
        return 1
    else:
        return 0

def decouple_rp():
    decouple.write(0,1)
    return decouple_status.read()

def couple_rp():
    decouple.write(0,0)
    return decouple_status.read()
```

Switching PR Configurations

```
In [64]: i = 1
hw_lat = []
sw_lat = []
overhead = []
for i in range(10):
    perf_rst.write(0,1)
    perf_rst.write(0,0)
    decouple_rp()

    start_time = time.time()
    reconfigure(i%2)
    stop_time = time.time()

    couple_rp()
    hw_del = (perf_counter.read()/(125*10**6))*10**3
    sw_del = (stop_time - start_time)*10**3

    hw_lat.append(hw_del)
    sw_lat.append(sw_del)
    overhead.append(sw_del - hw_del)

pcap_del = sum(hw_lat)/len(hw_lat)
print(f"avg. HW reconf time : {sum(hw_lat)/len(hw_lat)} ms")
print(f'avg. SW reconf time : {sum(sw_lat)/len(sw_lat)} mS')
print(f'avg. SW Overhead time : {sum(overhead)/len(overhead)} mS')
```

avg. HW reconf time : 1.1098792 ms
avg. SW reconf time : 74.07591342926025 mS
avg. SW Overhead time : 72.96603422926025 mS

Vivado Hardware Manager Flow (JTAG)

Reset Counter

```
In [65]: perf_rst.write(0,1)
perf_rst.write(0,0)
```

Use Hardware manager to send Bitstreams

```
In [68]: jtag_del = (perf_counter.read()/(125*10**6))*10**3
print(f"Latency with jtag : {jtag_del} ms")
```

Latency with jtag : 62.841168 ms

Comparing throughputs

```
In [69]: import os

pb_file_size = os.path.getsize(pr_files[0]["file"])

print(f"File size is {pb_file_size} bytes")
print(f"PCAP Throughput is {(pb_file_size/pcap_del)*10**-3} MB/s")
print(f"JTAG Throughput is {(pb_file_size/jtag_del)*10**-3} MB/s")
```

File size is 203308 bytes
PCAP Throughput is 183.1802956574013 MB/s
JTAG Throughput is 3.2352676831213576 MB/s

In []: